Analysis of Variance and Chi-square test

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Agenda

- Analysis of variance
  - Applications of statistical methods in engineering research and design
  - One way ANOVA
    - Total Variation
    - Variation within treatment
    - Variation between treatment
- Chi-square test
  - Goodness of Fit
  - Independence of Attributes

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Applications of statistical methods in engineering research and design

- Statistically based experimental design techniques are particularly useful in the engineering world for improving the performance of a manufacturing process.
- Most processes can be described in terms of several controllable variables, such as temperature, pressure, and feed rate. By using designed experiments, engineers can determine which subset of the process variables has the greatest influence on process performance.
- The results of such an experiment can lead to
  - Improved process yield
  - Reduced variability in the process and closer conformance to nominal or target requirements
  - Reduced design and development time
  - Reduced cost of operation

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- Designed experiments are usually employed sequentially. That is, the first experiment with a complex system (perhaps a manufacturing process) that has many controllable variables is often a screening experiment designed to determine which variables are most important.

- Every experiment involves a sequence of activities:

  - **Conjecture**—the original hypothesis that motivates the experiment.
  - **Experiment**—the test performed to investigate the conjecture.
  - **Analysis**—the statistical analysis of the data from the experiment.
  - **Conclusion**—what has been learned about the original conjecture from the experiment. Often the experiment will lead to a revised conjecture, and a new experiment, and so forth.

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Steps in Designing an Experiment

- **Define the Objective**: Clearly articulate the goal of the experiment and what you intend to investigate.
- Identify Factors and Responses: Determine which factors will be studied and how the responses will be measured.
- **Select a Design:** Choose an appropriate experimental design based on the number of factors, their levels, and the resources available.
- **Conduct the Experiment:** Execute the experiment according to the design, ensuring adherence to randomization, replication, and blocking principles.
- **Analyze Data:** Use statistical methods (like ANOVA) to analyze the collected data, determine significant factors and interactions, and draw conclusions.
- **Interpret Results:** Assess the implications of the findings for engineering practices, product development, or process improvements.
- **Refinement and Repetition:** Make adjustments based on findings and, if necessary, conduct further experiments to refine results.

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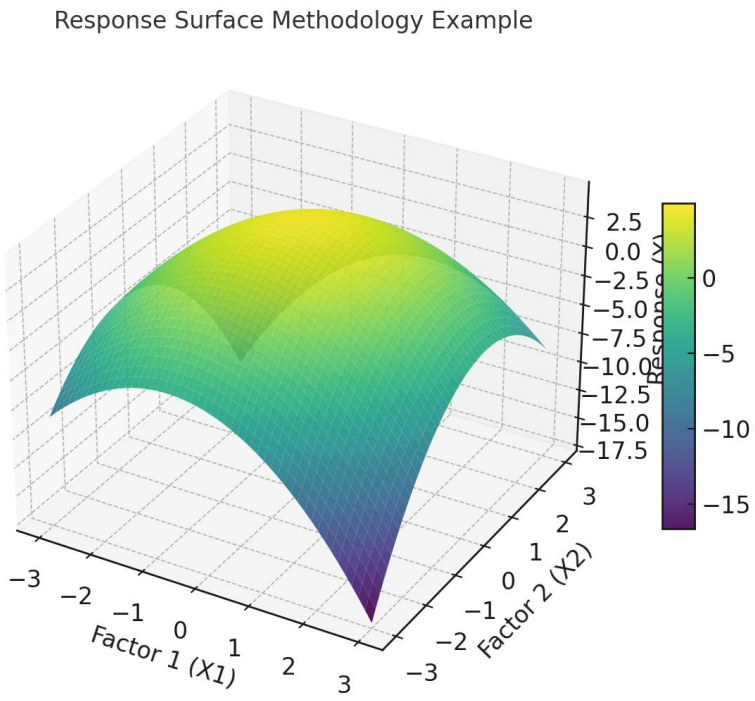
Response surface methodology (RSM)

- It is a **statistical and mathematical technique** for modeling and analyzing problems where multiple input variables influence an output (response).
- Uses **designed experiments** and **polynomial regression models** (often quadratic) to approximate the relationship.
- Instead of changing one variable at a time, RSM finds the **optimal combination of variables** efficiently.
- The "response surface" is a **visual representation** (2D/3D plot) showing how inputs affect the response. Figure shows how two factors (X1, X2) influence the response (Y), with a curved "surface" representing the optimization landscape.



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Where RSM is used?

- **Engineering & Manufacturing** → Process optimization (e.g., maximizing yield, minimizing defects).
- **Pharmaceuticals & Medicine** → Drug formulation optimization.
- **Agriculture** → Improving crop yields by adjusting multiple factors (fertilizer, irrigation, etc.).
- **Machine Learning/AI** → Hyperparameter tuning when searching for optimal model performance.

Experimental Design provides a structured way to plan experiments. Response Surface Methodology (RSM) builds models from these experiments to explore and optimize responses, while ANOVA is used within RSM to test the significance of factors and model terms.

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Question

Ryan is a production manager at an industry manufacturing alloy wires. They have 4 machines - A, B, C and D.

Ryan wants to study whether all the machines have equal efficiency based on the tensile strength of the alloy wire.

Is it possible to test his claim?

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Solution

The trivial solution is conducting multiple t tests. However performing multiple t tests has an effect on the type I error.

As the number of t-tests increases the probability of at least one type I error increases.

However, it is possible to test Ryans claim by using one way analysis of variance (one way ANOVA) where the probability of type I error does not change

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Multiple t tests and type I error

- For a true null hypothesis, the probability of not obtaining a significant result is 0.95 if the $\alpha = 0.05$

- Say you conduct the t-test twice, the probability of not obtaining one or more significant result is 0.95 x 0.95 = 0.9025

- Thus the probability of at least one type error is 1-0.9025 = 0.0975 (for two t-tests)

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Multiple t tests and type I error

Number of t tests	Probability of not obtaining one or more significant result	Probability of at least one type I error
3 t tests	0.95 x 0.95 x 0.95 = 0.857	0.143
4 t tests	0.95 x 0.95 x 0.95 x 0.95 = 0.815	0.185
5 t tests	0.95 x 0.95 x 0.95 x 0.95 x 0.95 = 0.774	0.226

As the number of tests increase the probability of at least one type error also increases

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ANOVA - History

- ANOVA was first introduced by Prof R. A. Fisher in 1920’s
- He developed ANOVA while dealing with agronomic data

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One way ANOVA

- A t-test is used when two unpaired data are compared
- To test the equality of population means for two or more unrelated samples ANOVA technique is used
- Each group is considered to be a treatment
- It is based on the F-distribution

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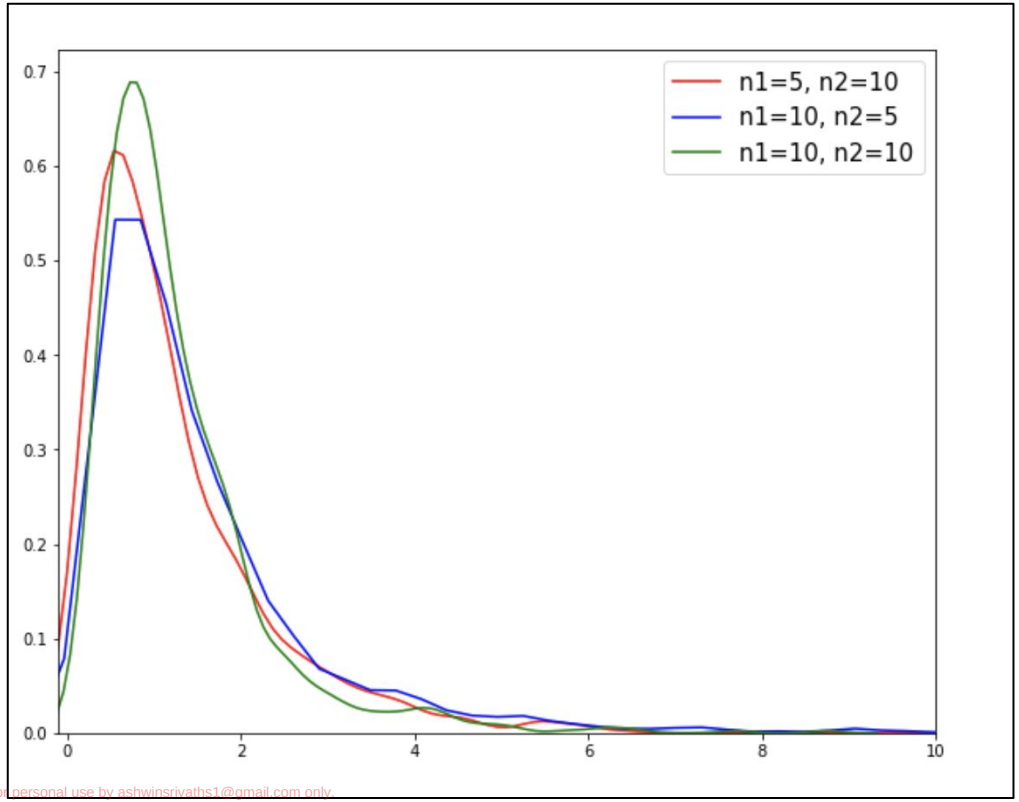
F distribution

- Let X be χ^2_m distribution and let Y be χ^2_n

- Then the ratio $\frac{X/m}{Y/n}$ follows F distribution with (m,n) df



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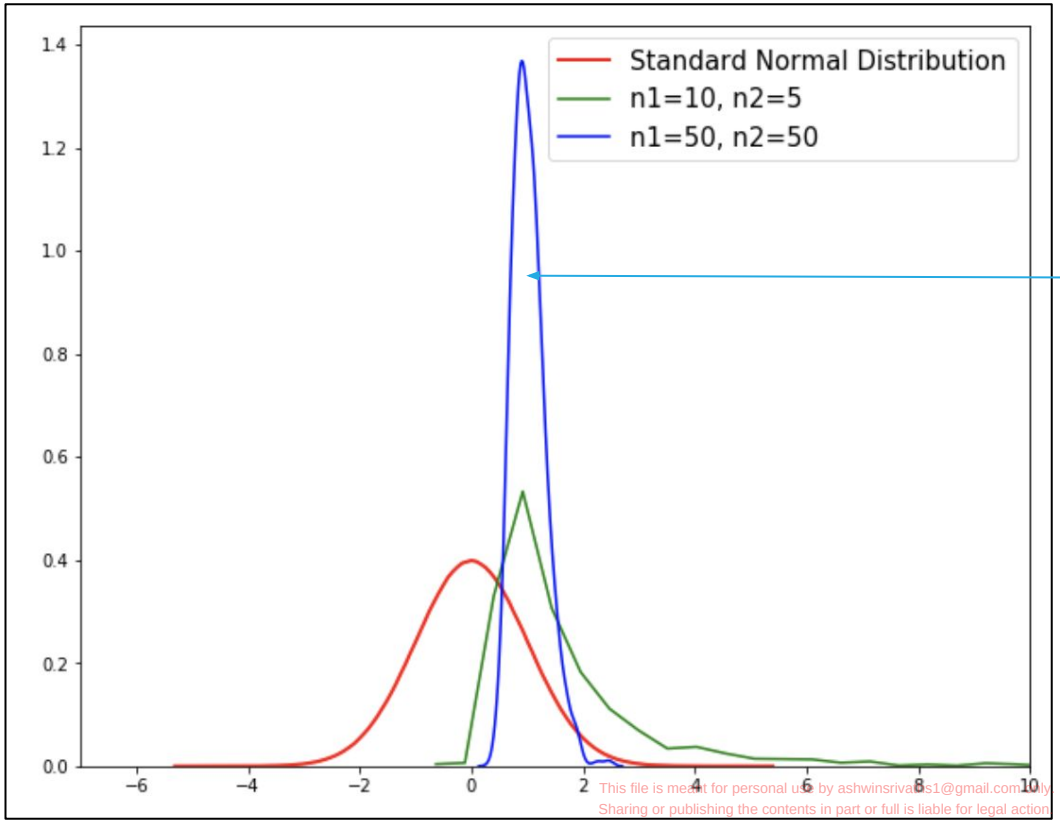
F distribution



As n_1 and n_2 become large the F distribution becomes symmetric

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One way ANOVA - assumption

- The samples should be independent
- Each sample should be from normally distributed population
- The population variance of the samples should be equal (homoscedastic)

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One way ANOVA

- The null hypothesis to be tested is

H_0: The averages of all treatments are same.
i.e. $\mu_1 = \mu_2 = \ldots = \mu_n$

Against

H_1: At the least one treatment has a different average

- Failing to reject H_0, implies that all treatments have the same average

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One way ANOVA

- Suppose Ryan collects data for tensile strength of wires produced by each machine
- It is said there are 4 treatments (t = 4)
- Each treatment has 5 observations (n_i = 5) where i = 1, 2, …, t
- Total number of observations is given by N

$$N = \sum_i^t n_i$$

A	B	C	D
68.7	62.7	55.9	80.7
75.4	68.5	56.1	70.3
70.9	63.1	57.3	80.9
79.1	62.2	59.2	85.4
78.2	60.3	50.1	82.3

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One way ANOVA

- Let μ_i (i=1, 2, …, t) denote the average strength due to each machine

- For our example, t = 4

- The test hypothesis can be written as

H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4$ Against H_1: At least μ_i is different

A	B	C	D
68.7	62.7	55.9	80.7
75.4	68.5	56.1	70.3
70.9	63.1	57.3	80.9
79.1	62.2	59.2	85.4
78.2	60.3	50.1	82.3

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One way ANOVA

- In one way ANOVA, the entire population variance is split into two component
  - Variation within treatment
  - Variation between treatment
- Total variation = Within treatment variation + Between treatment variation

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Total variation

- It is the total sum of squares (TSS)
- Let x_{ij} be the observations in the ith treatment and j^{th} row
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The total variation is given by

$$TSS = \sum_i^t \sum_j^{n_i} (x_{ij} - \bar{x}_{..})^2$$

Summation over all treatments

Summation over all observation in the treatment

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Within treatment variation

- It is the treatment sum of squares (TrSS)
- Let x_i be the observations in the ith treatment with n_i in observation in each treatment and $\bar{x}_{i.}$ is the mean over i^{th} treatment
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The treatment variation is given by

$$TrSS = \sum_i^t \sum_j^{n_i} n_i(\bar{x}_{i.} - \bar{x}_{..})^2$$

Summation over all treatments

Summation over all observation in the treatment

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Between treatment variation

- It is the error sum of squares (ESS)
- Let $x_{i.}$ be the observations in the ith treatment and $\bar{x}_{.j}$ is the mean over j^{th} row
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The error sum of squares is given by

$$ESS = \sum_i^t \sum_j^{n_i} (x_{ij} - \bar{x}_{i.})^2$$

Summation over all treatments

Summation over all observation in the treatment

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Error sum of squares

During problem solving, the error sum of squares is obtained as:

$$\text{ESS} = \text{TSS} - \text{TrSS}$$

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One way ANOVA

- The test statistic is given by

$$\text{F-ratio} = \frac{\frac{TrSS}{df_{Tr}}}{\frac{ESS}{df_e}} = \frac{\text{MTrSS}}{\text{MESS}}$$

Mean Treatment Sum of Squares

Mean Error Sum of Squares

- Under H_0, the test statistic follows F-distribution with (df_{Tr}, df_e) degrees of freedom

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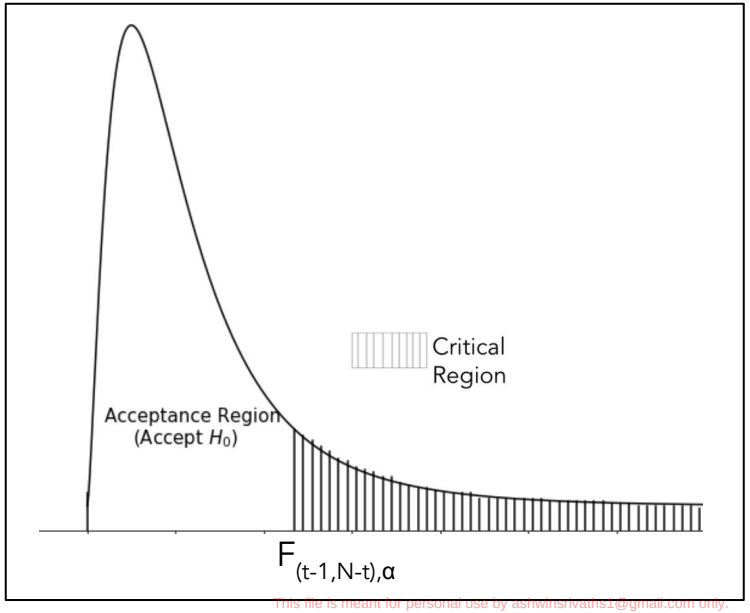
One way ANOVA

Decision Rule: If $F_{cal} \geq F_{(t-1,N-t),\alpha}$ or p-value $\leq \alpha$, then we reject H_0 at α% level of significance



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One way ANOVA

To ease the entire computational process, an ANOVA table is prepared as follows:

Source of variation	Degrees of freedom	Sum of Squares	Mean Sum of Squares	F-ratio
Treatment	t-1	TrSS	s^2_t	$\frac{s_t^2}{s_e^2}$
Error	N-t	ESS	s^2_e	
Total	N-1	TSS		

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One way ANOVA - procedure

1. State the hypothesis to be tested
2. Compute the sum of squares
   a. The total sum of squares, TSS = $\sum_{j=1}^{t}\sum_{i=1}^{ni} (x_{ij} - \bar{x}_{..})^2$
   b. The treatment sum of squares TrSS = $\sum_{j=1}^{t}\sum_{i=1}^{ni} n_i(x_{ij} - \bar{x}_{i.})^2$
   c. The Error sum of squares, ESS = TSS - TrSS
3. Compute mean sum of squares
   a. s^2_t = Mean group sum of squares (MTrSS) = TrSS/(t-1)
   b. s^2_e = Mean error sum of squares (MESS) = ESS/(N-t)

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One way ANOVA - procedure

4. Compute the F-ratio

$$\text{F-ratio} = \frac{\text{MTrSS}}{\text{MESS}} = \frac{s_t^2}{s_e^2}$$

4. Prepare the ANOVA table

5. Write the decision and conclusion accordingly

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One way ANOVA

Question:

Ryan is a production manager at an industry manufacturing alloy seals. They have 4 machines - A, B, C and D. Ryan wants to study whether all the machines have equal efficiency.

Ryan collects data of tensile strength (in N/m^2) from all the 4 machines as given.

Test at 5% level of significance.

A	B	C	D
68.7	62.7	55.9	80.7
75.4	68.5	56.1	70.3
70.9	63.1	57.3	80.9
79.1	62.2	59.2	85.4
78.2	60.3	50.1	82.3

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One way ANOVA

Solution:

Ryan is a production manager at an industry manufacturing alloy seals. They have 4 machines - A, B, C and D

Let μ_1 be the average tensile strength due to machine A
μ_2 be the average tensile strength due to machine B
μ_3 be the average tensile strength due to machine C
μ_4 be the average tensile strength due to machine D

To test, H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4$ Against H_1: At least one μ_i is different (i=1, 2, 3, 4)

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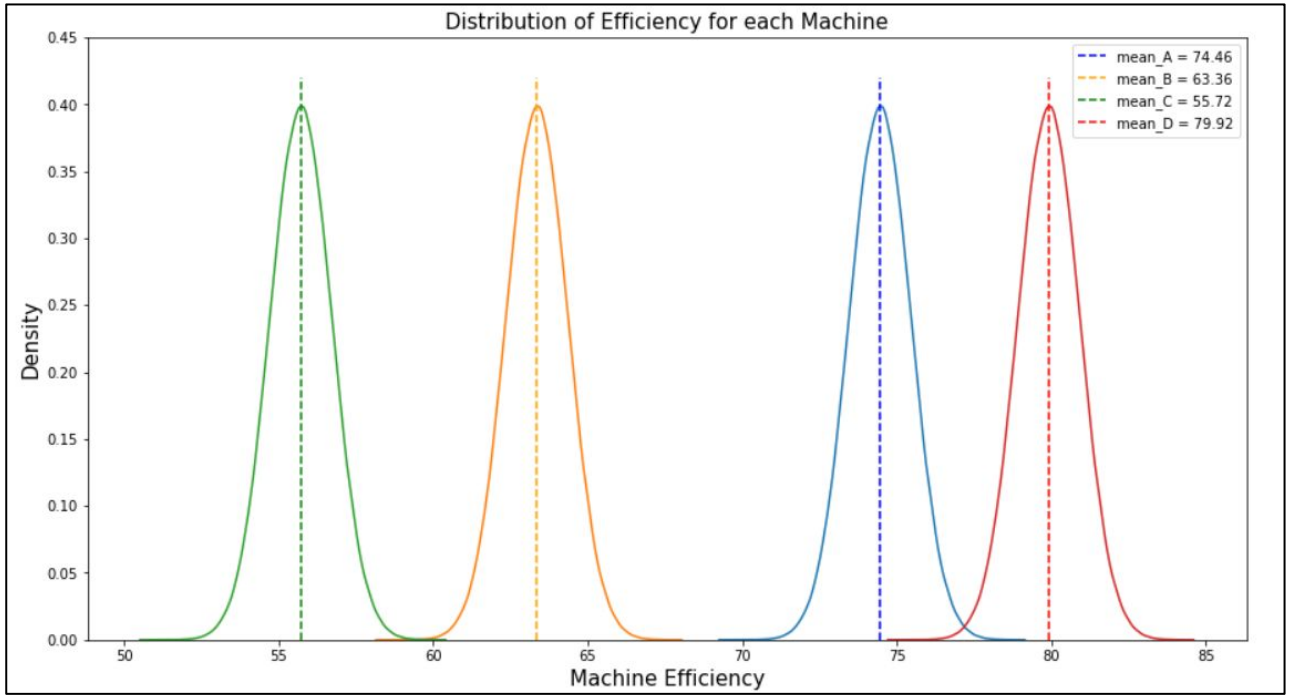
One way ANOVA

The plot shows the difference between the average efficiency for each machine, which indicates the rejection of H_0.



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One way ANOVA

Solution:

The grand mean:

$$\bar{x}_{..} = \frac{68.7+62.7+...+50.1+82.3}{20} = 68.365 \ \mathrm{N/m}^2$$

The total sum of squares:

$$TSS = \sum_i^t \sum_j^{n_i} (x_{ij} - \bar{x}_{..})^2$$

$$= (68.7 - 69.115)^2 + \ldots + (81.12 - 69.115)^2$$

$$= 2074.1255 \ (\mathrm{N/m}^2)^2$$

A	B	C	D
68.7	62.7	55.9	80.7
75.4	68.5	56.1	70.3
70.9	63.1	57.3	80.9
79.1	62.2	59.2	85.4
78.2	60.3	50.1	82.3

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One way ANOVA

Solution:

The treatment sum of squares is

$$TrSS = \sum_i^t \sum_j^{n_i} n_i(\overline{x}_{i.} - \overline{x}_{..})^2$$

$$= 5(74.46 - 68.365)^2 + \ldots + 5(79.92 - 68.365)^2$$

$$= 1778.0655 \ (\mathrm{N/m}^2)^2$$

	A	B	C	D
	68.7	62.7	55.9	80.7
	75.4	68.5	56.1	70.3
	70.9	63.1	57.3	80.9
	79.1	62.2	59.2	85.4
	78.2	60.3	50.1	82.3
$\sum x_i$	372.3	316.8	278.6	399.6
$\overline{x}_{i.}$	74.46	63.36	55.72	79.92

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One way ANOVA

Solution:

The error sum of squares can also be calculated as

$$ESS = \sum_i^t \sum_j^{n_i} (x_{ij} - \bar{x}_{i.})^2$$

$$= (68.7 - 74.46)^2 + \ldots + (82.3 - 79.92)^2$$

$$= 296.06$$

	A	B	C	D
	68.7	62.7	55.9	80.7
	75.4	64.5	56.1	80.3
	70.9	63.1	57.3	80.9
	79.1	59.2	55.2	81.4
	78.2	60.3	50.1	82.3
$\bar{x}_{i.}$	74.46	63.36	55.72	79.92

Or can be obtained as $\mathrm{ESS} = \mathrm{TSS} - \mathrm{TrSS} = 296.06$

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One way ANOVA

Solution:

Source of variation	Degrees of freedom	Sum of Squares	Mean Sum of Squares	F-ratio
Treatment	t-1 = 4-1 =3	TrSS = 1778.0655	$s_t^2 = \frac{21778.0655}{3} = 592.6885$	$\frac{s_t^2}{s_e^2} = 32.031$
Error	N-t = 20-4 =16	ESS = 296.06	$s_e^2 = \frac{269.06}{16} = 18.50375$	
Total	N-1 = 20 -1 =19	TSS = 2241.5255		

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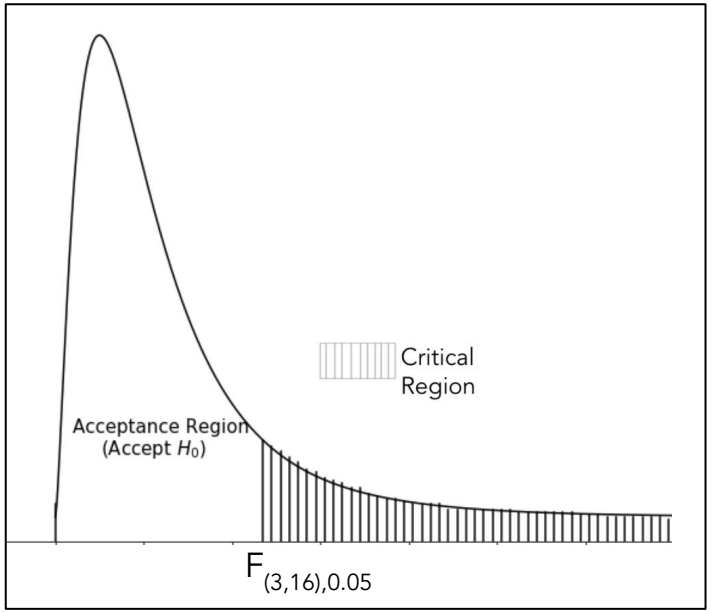
One way ANOVA

Solution:

From the F-table we have $F_{(3,16),0.05} = 3.24$

Since $3.24 < 32.03$, we reject H_0.



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One way ANOVA

Python solution:

```
# perform one-way ANOVA
# pass the given data using the dataframe 'df_machine'
test_stat, p_val = stats.f_oneway(df_machine[df_machine['machine'] == 'machine_A']['strength'],
                                  df_machine[df_machine['machine'] == 'machine_B']['strength'],
                                  df_machine[df_machine['machine'] == 'machine_C']['strength'],
                                  df_machine[df_machine['machine'] == 'machine_D']['strength'])

# print the test statistic and p-value
print('Test statistic:', test_stat)
print('p_value:', p_val)
```

```
Test statistic: 32.03072350199285
p_value: 5.375613532781072e-07
```

As p-value < 0.05, we reject H_0.

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PLEASE!! NOTE

One way ANOVA can be said to check the effect of a nominal variable over a numerical variable.

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Tests for Categorical Data

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Tests for categorical data

- The collected data is at times best represented by categories
- These categories are summarized by their frequency of occurrence. It may be of interest whether this frequency is equal to the expectation/claim
- It may also be of interest to know whether the categories are statistically independent

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Chi-square tests

- These interests are tested by non-parametric ways
- Tests based on the chi-square distribution are used
- The chi-square tests are used to test:
  - The goodness of fit
  - The independence of two attributes
- Chi-square tests are also used to test for population variance

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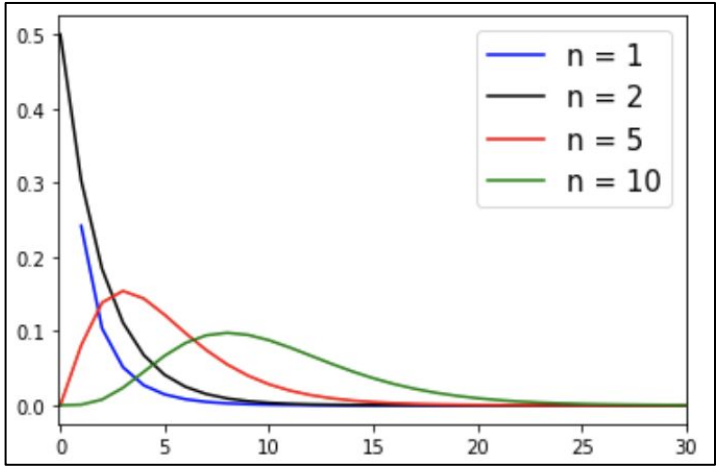
Chi-square distribution

- Chi-square distribution is sum of squared of standard normals

  Let $X_1, X_2, \ldots, X_n$ be n standard normal variates,

  then $Y = X^2_1 + X^2_2 + \ldots + X^2_n$,

  Y follows χ^2 distribution with *n* degrees of freedom.



- The mean of the distribution is *n* and its variance is *2n*

- The distribution is positively skewed

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χ^2 Test for Goodness of Fit

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Chi-square test for goodness of fit

At an emporium, the manager is interested in knowing age group which visits the mall during the day. He defines categories as - children (age < 13), teenagers (13 ≤ age < 20), adults (20 ≤ age < 55) and senior citizens (55 ≤ age). Moreover, he wishes to plan his inventory of goods accordingly.

He claims that out of all the people who visited 5% are children, 38% are teenagers, 2% are senior citizens are remaining are adults.

Can the owner verify the managers claim?

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Chi-square test for goodness of fit

- The hypothesis to test whether the data fits the a specified distribution

H_0: There is no difference between observed frequencies and expected frequencies

Against

H_1: There is difference between observed frequencies and expected frequencies

- Failing to reject H_0, implies that there is no difference between observed frequencies and expected frequencies

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Chi-square test for goodness of fit

- The test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{O_i^2}{e_i} - N$$

observed frequency

total number of observations

estimated frequency

- Under H_0, the test statistics follows χ^2 distribution with k-p-1 d.f

  where *k*: number of class frequencies

  p: number of parameter estimated for fitting

Ref: Test statistic for goodness of fit (A.1)

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Chi-square test for goodness of fit

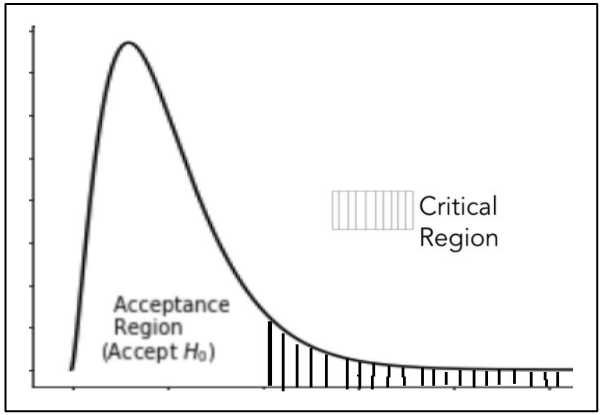
Decision Rule:

Reject H_0 if $\chi^2_{calc} \geq \chi^2_{k-p-1,\alpha}$

or

Reject H_0 if p-value $\leq \alpha$

Where, α is the level of significance (l.o.s.)



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Test for goodness of fit

Question:

At an emporium, the manager is interested in knowing age group which visits the mall during the day. He defines categories as - children, teenagers, adults and senior citizens. He plans to have his inventory of goods accordingly. He claims that out of all the people who visited 5% are children, 38% are teenagers, 2% are senior citizens are remaining are adults.

From a sample of 180 people it was seen that 25 were children, 50 were teenagers, 90 were adults and 15 were senior citizens

Test the manager's claim at 95% confidence level.

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Test for goodness of fit

Solution:

We can tabulate the given data as follows:

	Manager Claimed Frequency	The frequency expected from 180 customers (e_i)	The frequency observed from 180 customers (O_i)
Children	5%	0.05 x 180 = 9	25
Teenagers	38%	0.38 x 180 = 68.4 ≅ 68	50
Adults	55%	0.55 x 180 = 99	90
Senior Citizens	2%	0.02 x 180 = 3.6 ≅ 4	15

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Test for goodness of fit

Solution:

To test, H_0: The managers claim is correct Against H_1: The managers claim is false

The test statistic is $\chi^2 = \sum_{i=1}^{k} \frac{O_i^2}{e_i} - N$

$$= [\frac{25^2}{9} + \frac{50^2}{68} + \frac{90^2}{99} + \frac{15^2}{4}] - 180$$

$$= 64.27$$

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Test for goodness of fit

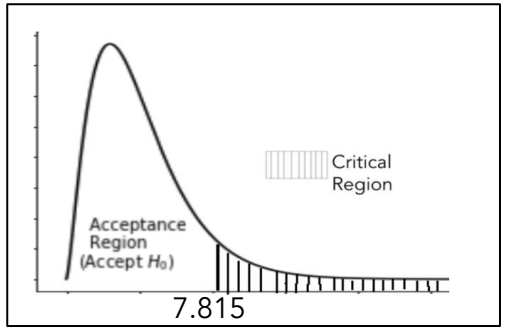
Solution:

Here there are 4 class frequencies, i.e k = 4. Since no parameter was calculated p = 0

From the statistical table for χ^2 distribution, $\chi^2_{k-p-1,\alpha} = \chi^2_{3,0.05} = 7.815$

The test statistic $\chi^2_{calc} = 64.27$

Since $\chi^2_{calc} > \chi^2_{k-p-1,\alpha}$, reject H_0.



The managers claim is false, his claim is different than what was observed from the data

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Test for goodness of fit

Python solution:

```python
# given observed values
observed_value = [25, 50, 90, 15]

# expected count
exp_count = [0.05, 0.38, 0.55, 0.02]

# calculate the expected values for each category
# expected_value = (np.array(exp_count) * 180)
expected_value = [9, 68, 99, 4]

# use the 'chisquare()' to perform the goodness of fit test
# the function returns the test statistic value and corresponding p-value
# pass the observed values to the parameter, 'f_obs'
# pass the expected values to the parameter, 'f_exp'
stat, p_value = chisquare(f_obs = observed_value, f_exp = expected_value)

print('Test statistic:', stat)
print('p-value:', p_value)
```

```
Test statistic: 64.2773321449792
p-value: 7.160266387019384e-14
```

As p-value < 0.05, we reject H_0.

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PLEASE!! NOTE

- If the expected frequencies $e_i \geq 5$ and the total frequencies are large (≥ 50) the test can be used

- If $e_i < 5$, the class is merged with the neighbouring class for observed and expected frequencies until the it becomes ≥ 5

- It is not applicable for testing the goodness of fit of a straight line or any curve (exponential curve, second degree curve)

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χ^2 Test for Independence of Attributes

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Chi-square test for independence of attributes

- The hypothesis to test independence of attributes

  H_0: The attributes are independent against H_1: The attributes are dependent

- Failing to reject H_0, implies that the attributes are independent

- Decision rule: Reject H_0 at α l.o.s if $\chi^2_{(r-1)(s-1)} \geq \chi^2_{(r-1)(s-1);\alpha}$

  or

  Reject H_0 if p-value $\leq \alpha$

  Where, α is the level of significance (l.o.s.)

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Chi-square test for independence of attributes

- The test statistic is given by

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{O_{ij}^2}{e_{ij}} - N$$

observed frequency in i^{th} row and j^{th} column

total number of observations

estimated frequency in i^{th} row and j^{th} column

- Under H_0, the test statistics follows χ^2 distribution with (r-1)(c-1) d.f where *r* levels for a category and *c* levels for another category

Ref: Test statistic for independence of attributes (A.2)

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Test for independence of attributes

Question:

A study was conducted to test the effect of the malaria parasite - plasmodium falciparum - on heterozygous and homozygous humans. The vaccine was given to a cohort of 252 humans. Test whether the heterozygous humans are better protected than homozygous.

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum
Heterozygous humans	93	51
Homozygous humans	68	40

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Test for independence of attributes

Solution:

Let X: The zygote type - Homozygous or Heterozygous
Y: Whether infected or not with malaria parasite

Here X and Y are two attributes.

To test, H_0: The attributes are independent Against H_1: The attributes are dependent

Here there are 2 rows and 2 columns.

Let us computed the expected frequency.

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Test for independence of attributes

Solution:

In order to compute the expected frequency, first compute the total of the each column and row.

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	93	51	144
Homozygous humans	68	40	108
Total	161	91	252

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Test for independence of attributes

Solution:

The expected frequencies are computed as

$$e_{ij} = \frac{\text{Total}_{\text{row}} \times \text{Total}_{\text{column}}}{N}$$

$$e_{11} = \frac{144 \times 161}{252} = 92$$

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	$\frac{144 \times 161}{252} = 92$		144
Homozygous humans			108
Total	161	91	252

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Test for independence of attributes

Solution:

Similarly compute the expected frequencies for other classes

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	$\frac{144 \times 161}{252} = 92$	$\frac{144 \times 91}{252} = 52$	144
Homozygous humans	$\frac{108 \times 161}{252} = 69$	$\frac{108 \times 91}{252} = 39$	108
Total	161	91	252

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Test for independence of attributes

Solution:

The observed frequency (O_{ij})

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	93	51	144
Homozygous humans	68	40	108
Total	161	91	252

The expected frequency (e_{ij})

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	92	52	144
Homozygous humans	69	39	108
Total	161	91	252

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Test for independence of attributes

Solution:

The test statistic is computed as

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{O_{ij}^2}{e_{ij}} - N$$

$$= \frac{93^2}{92} + \frac{51^2}{52} + \frac{68^2}{69} + \frac{40^2}{39} - 252$$

$$= 0.070$$

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Test for independence of attributes

Solution:

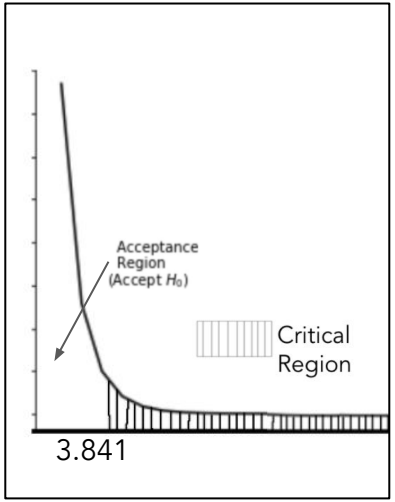
Here there are 2 levels of one attribute and 2 levels of another.
Thus the degrees of freedom are (r-1)(c-1) = (2-1)(2-1) = 1

From the statistical table for χ^2 distribution, $\chi^2_{(r-1)(s-1),\alpha} = \chi^2_{1,0.05} = 3.841$

The test statistic $\chi^2_{calc} = 0.070$

Since $\chi^2_{calc} < \chi^2_{k-p-1,\alpha}$, we fail to reject H_0.

The attributes are independent.



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Test for independence of attributes

Python solution:

```
# use the 'chi2_contingency()' to check the independence of variables
# pass the observed values to the parameter, 'observed'
# 'correction = False' will not apply the Yates' correction
test_stat, p, dof, expected_value = chi2_contingency(observed = observed_value, correction = False)

# print the output
print("Test statistic:", test_stat)
print("p-value:", p)
```

```
Test statistic: 0.07023411371237459
p-value: 0.790996215494177
```

As p-value > 0.05, we fail to reject H_0.

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Independence of attributes

Question:

A psychologist wants study whether the happiness quotient of children in the house is related to the family income. He collects data of 1300 children is there enough evidence to claim that they are related.

	Low income	Moderate income	High income
Happy	245	354	243
Unsatisfied	98	220	140

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PLEASE!! NOTE

Tests based on Chi-squared distribution for categorical data are one tailed tests.

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